ML LAB Mini Project

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COVID-19 Clinical Severity & Risk Estimation

1. Problem Statement

Stanford CS229 report by Zhan & Li (2020)

Problem Statement No.: 12

2. Dataset

Source: COVID-19 Mexico Patient Health Dataset (Kaggle, from the Mexican Ministry of Health surveillance system).

- **11 input features:** Age (standardised with **StandardScaler**) plus binary flags for gender, pneumonia, diabetes, COPD, asthma, immunosuppression, hypertension, cardiovascular disease, smoking and obesity. All are assumed known at admission, as in the paper.
- **3 binary targets:** ICU admission, intubation and death.

Outcome	Positives (total)	Prevalence	Positives in test set
ICU admission	987	0.04	198 / 4,693
Intubation	991	0.04	195 / 4,693
Death	2154	0.09	427 / 4,693

3. Approach

Because only about 4% of patients need ICU care or intubation, a model that always predicts “no” would score about 95.8% accuracy while missing every critical patient. Accuracy is therefore a poor metric here. In triage, a false negative (a deteriorating patient not flagged) is far more costly than a false positive (extra monitoring for a stable patient), so prioritised recall is necessary, as the paper does, and tracked ROC-AUC and PR-AUC alongside it.

- **Imbalance handling:** class_weight="balanced" in every model.
- **Models compared:** Logistic Regression, SVM (RBF kernel), Decision Tree (max_depth = 6, min_samples_leaf = 50) and a tree-ensemble model, each trained as three separate binary classifiers (one per outcome) on identical features and splits.
- **Selection criterion:** discrimination (ROC-AUC), recall on the positive class, simplicity and interpretability.

4. Implementation Overview

The work is in a Python/scikit-learn notebook (model_training.ipynb). Steps: (1) load patient.csv and filter to positive cases; (2) rename and clean columns and binarise flags; (3) build the 11-feature input table and three target tables; (4) fit one scikit-learn Pipeline per outcome, a ColumnTransformer that standardises Age and passes binary flags through, followed by the classifier; (5) evaluate on the held-out set with classification reports, confusion matrices, ROC and precision-recall curves; (6) export the three logistic regression pipelines with joblib to Models/. A Streamlit app (demo.py) loads these and takes age, gender and conditions as input. It returns ICU, intubation and death risk percentages with “Flagged: higher risk” or “Not flagged” badges.

Test-set results (positive class)

Model	ICU Rec.	ICU AUC	Int. Rec.	Int. AUC	Death Rec.	Death AUC	Death Prec.
Logistic Regression	0.84	0.83	0.94	0.88	0.8	0.85	0.25
SVM	0.82	0.8	0.92	0.85	0.83	0.81	0.23
Decision Tree	0.83	0.83	0.9	0.85	0.81	0.84	0.23
Random Forest*	0.83	0.83	0.9	0.85	0.81	0.84	0.23

5. Conclusions

- **Useful as a screening filter.** T**he models catch 80–94%** of severe cases and correctly clear roughly **74% of patients**, with a negative predictive value of 0.97–1.00. A positive flag is only a weak signal (precision 0.12–0.25), **which is the accepted cost of high recall in triage.**
- **Logistic regression is the deployed model.** Differences between algorithms are small (ROC-AUC 0.80–0.88). It has the best ROC-AUC on intubation and death, ties on ICU, and is the simplest to explain and the easiest to turn into probabilities.
- **Pneumonia dominates risk.** Its coefficient is the largest for all three outcomes (about 44× higher odds of intubation). Age is the next driver of mortality, followed by obesity, diabetes and immunosuppression. Death is the most predictable outcome.
- **Limits and next steps.** Four different algorithms converge to similar scores, so the ceiling is set by the information in the features, not the model. Adding CT radiomic features, as in the Stanford paper, is the most likely route to better precision.